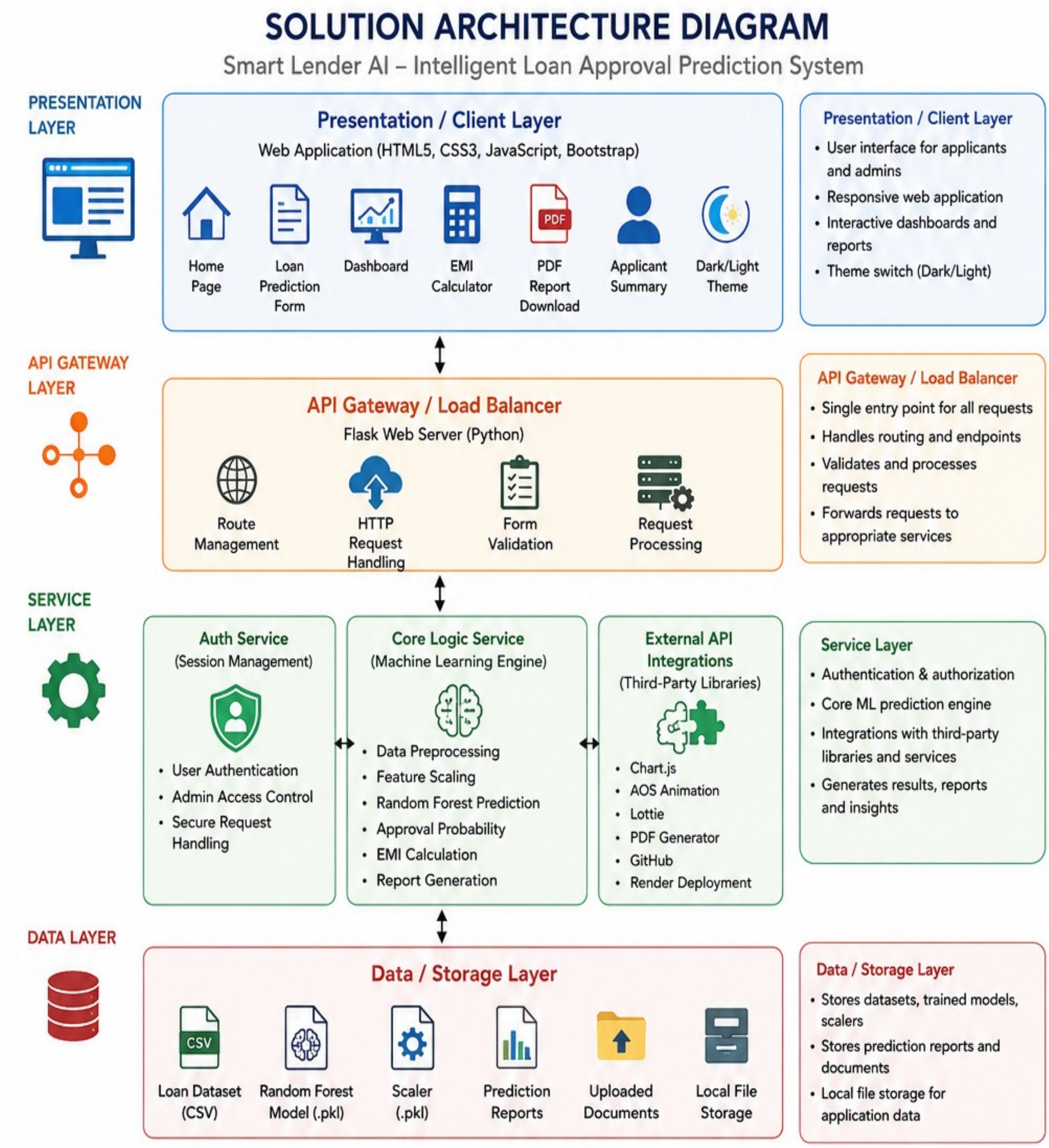
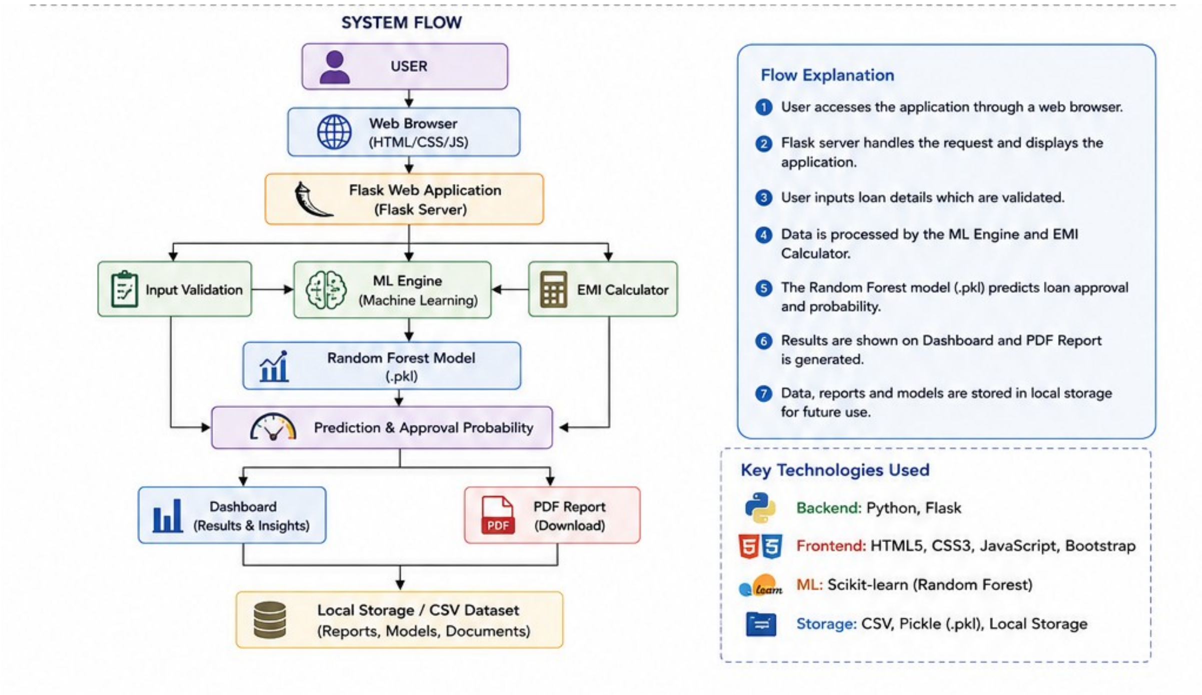
Solution Architecture

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:



System Flow



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the graphical user interface where users enter loan details, view predictions, dashboard analytics, EMI calculations, and download reports.	HTML5, CSS3, JavaScript, Bootstrap, Chart.js
API Gateway	Receives user requests, validates input data, manages routing, communicates with the Machine Learning engine, and returns prediction results.	Python, Flask
Core Logic Service *(instead of "Microservices")*	Processes applicant information, performs feature scaling, executes the Random Forest model, calculates EMI, generates approval probability, and prepares reports.	Scikit-learn, NumPy, Pandas, Pickle
Database / Storage	Stores the loan dataset, trained Machine Learning model, scaler, generated reports, and uploaded files required for prediction and analysis.	CSV Dataset, Pickle (.pkl), Local Storage